

NetraSetu — Technical Documentation

PS 26038 — Explainable AI for Diabetic Retinopathy Screening in Rural India Team: Tanuj (lead, ML), Saad (backend, MATLAB/Simulink), Kankshi, Parth, Vedant — "Game Of Codes"

This document explains what the system is, how its parts fit together, and what state each part is actually in today. It is built from `docs/system-design-v4.md` (the team's locked design document) reconciled against `docs/STALE_CLAIMS_AUDIT.md` (a fresh, code-verified audit of what in v4 is still accurate vs. now resolved) and this project's own final integration-testing pass — so where v4 describes something as unbuilt or broken and it has since been fixed and verified live, this document says so, with the same file:line discipline v4 itself uses. Full ML numbers live in `docs/ML_BENCHMARKS.md`; this document explains *why* the system is built the way it is and *what* each piece does.

1. The Problem, and Why the Architecture Looks the Way It Does

India has roughly one ophthalmologist per 100,000 rural people, and diabetic retinopathy affects about 18% of the country's 77M+ diabetic adults. Early screening prevents 90% of DR-related blindness, but there aren't enough specialists to screen at population scale. The actual task isn't "build an AI that detects DR" — it's "let a non-specialist technician at a rural Primary Health Centre (PHC) do what only a specialist could do before, and let that specialist trust and act on the AI's output in seconds." Every architectural choice below traces back to that sentence.

Edge/cloud split. Rural PHCs have unreliable or absent internet and modest hardware. The problem statement's accuracy bar is hard enough to hit with a full-capacity model — compressing it to survive on edge hardware risks losing exactly the accuracy the system is graded on. The only thing that genuinely benefits from running locally is the image-quality check, because a technician needs an instant retake decision before the patient leaves the chair. So: **local does quality gating only; central does all grading.** This applies identically to both front-ends.

Capture source. A phone's bare camera cannot produce a usable fundus photograph — this is an optical limitation (purpose-built optics are needed to see through the pupil), not something software fixes. Both front-ends require a dedicated fundus camera as the only way an image enters the system; the mobile app's "capture" step is a gallery import of an image taken on that same camera, not a second, competing capture method. Nothing in either front-end calls an external, unversioned inference endpoint — an earlier prototype iteration of both the desktop capture screen and the mobile app did exactly that, and neither behavior carries forward into the current system.

Human-in-the-loop, always. The AI never tells a patient they have a disease directly. Every "probably has DR" outcome — real-time or delayed — is confirmed by an ophthalmologist before it becomes an SMS to the patient.

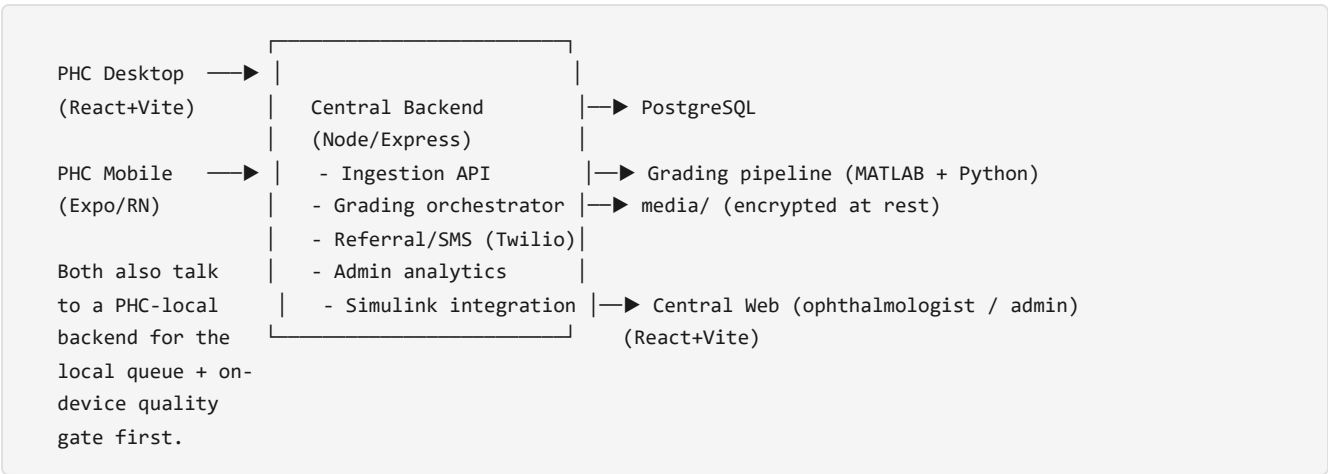
Two front-ends, one spec. A desktop web app and an Expo mobile app both implement the same PHC technician workflow against the exact same central API contract — neither has its own backend, response shape, or capture-time inference call. They differ in exactly one respect: how the image gets onto the device (camera SDK vs. gallery import of an image from the same camera). Everything after that point — quality gate, questionnaires, sync, identity scheme, definition of "synced" — is the same spec, met the same way on both.

No result ever substitutes for a failure. A network failure, a timeout, and a working result must be distinguishable on screen. A failed request either surfaces as an error or genuinely queues for retry — never a plausible-looking fabricated result. This is a system-wide rule, not one screen's convention, and it is why mock data is only ever shown behind an explicit, visible "DEMO DATA" banner (`DATA_MODE=mock`), never as an automatic fallback on error.

2. Actors

Actor	Touches
PHC Technician	Either local front-end (desktop or mobile)
Patient	Receives an in-person result and/or SMS; gives verbal consent, timestamped at registration
Ophthalmologist	Central web app — review queue, case detail, decision controls
District Health Administrator	Central web app — dashboard, system health, referral tracker, resource recommendations
ASHA / Community Health Worker	Referral follow-up, assigned via the admin interface

3. High-Level Architecture



Where MATLAB lives: the quality gate, preprocessing, camera calibration, segmentation orchestration, the rule engine, calibration math, explainability, and the Simulink resource model are MATLAB/toolbox territory. The UI, database, and API layers are not. Branch A (the CNN classifier) is trained in PyTorch, exported to ONNX, and imported into MATLAB's Deep Learning Toolbox so the actual inference call runs inside a persistent MATLAB session the backend calls into per case — a real, verified capability, not a demo trick, though it assumes a MATLAB license on whichever machine runs it (see §10, Honest Limitations).

No Redis. The grading queue runs in-process in the central backend.

4. Local PHC Application (Both Front-Ends)

4.1 Screens

Screen	What it does
Patient Registration	New-patient form; fuzzy duplicate check by name/age/phone before a new patient ID is minted; verbal-consent checkbox, timestamped
Capture	Desktop: camera trigger. Mobile: gallery import of an image from the same dedicated camera. Both tag the capture left/right eye before anything else touches it
Quality Result	Pass / borderline / fail with a specific reason (blur, dark, poor field of view, glare, motion, eyelash occlusion, etc.); a real quality score is always shown when one exists, "not available" when it doesn't — never a fabricated number
Patient Symptom/Risk Questionnaire	Diabetes duration, glycemic control, BP, pregnancy status, symptom toggles — no free text, no skip
Capture Metadata Questionnaire	Device, pupil status, lighting, observed issues — tap-only, under 45 seconds
Local Queue	Captured → Quality-Passed → Synced → Result-Pending → Result-Delivered, with a storage-pressure indicator
Sync Status	Online/offline indicator, pending count; "synced" means the central backend actually accepted the case, never merely that a request returned

4.2 Local Backend

Service	Responsibility
Capture Handler	Desktop: camera driver interface. Mobile: gallery-import step. Either way, tags the image by eye
Quality-Gate Engine	Desktop: MATLAB (<code>matlab -batch</code> , with a compiled-exe tier and a pure-JS fallback tier if MATLAB is unavailable — deliberate resilience, not a gap). Mobile: the same JS reimplementation as the desktop's fallback tier, ported once, not reinvented, so both front-ends apply identical rules
Local API	Wraps Capture Handler + Quality-Gate Engine, reads/writes the local DB
Sync Manager	Prioritizes by urgency tier then age; sends a lightweight case-summary packet ahead of the full image on thin connectivity; full images go through resumable, chunked upload; every submission carries the local <code>capture_id</code> as an idempotency key so a retried upload never creates a duplicate case

4.3 What Differs on Mobile, and Only This

- **Local persistence** is a real embedded database (`expo-sqlite`), not a single serialized blob — per-record queries and safe partial writes under an extended outage with dozens of cases queued.
- **Identity** uses the exact same collision-safe ID scheme as desktop (PHC code + timestamp + random suffix), not a second format that could silently drift apart from it.
- **Both questionnaires apply in full** — no skip option, same field set, on either front-end.
- A dedicated on-device **fundus-lens capture mode** exists (`mobile_lens` camera preset) in addition to gallery import, with its own lens-specific capture guidance (attach the lens, dim the room, find the red reflex, align the optic disc in the guide circle).

5. Central System

5.1 Ophthalmologist Interface

Screen	Components
Review Queue	Sorted by urgency tier first (least-confident cases surface first, not most severe); patient ref, eye, PHC, capture time, both branches' grades, confidence tier, claim status
Case Detail	Fundus image with a toggleable Grad-CAM overlay; four-category lesion-evidence panel; both grading branches side by side with a disagreement flag; calibrated confidence and conformal tier; an automatic review-duration timer; a "show technical/engine details" toggle (collapsed by default, auto-opened only if a fallback engine actually ran)
Decision Controls	Opening a case claims it (a second reviewer sees who holds it and cannot submit a conflicting decision). Confirm, or Override with a structured reason + optional free text + corrected grade. On branch disagreement, "Confirm" is not available — the reviewer must pick an explicit final grade
Case History	Per-patient longitudinal view — grade-over-time trend across visits, per eye, including prior lesion counts and prior review/referral status

5.2 District Admin Interface

Screen	Components
Dashboard	Cases today/week/total, average review time, override rate, average confidence, cases by PHC
System Health	One consolidated view: PHC sites gone silent, grading jobs stuck, the MATLAB session's own health, referable cases sitting unreviewed past a threshold — plain-language summaries by default, raw technical detail behind a toggle
Referral Tracker	referred → contacted → attended / lost, assigned-worker field, automatic manual-follow-up state on SMS delivery failure
Resource Recommendations	Staffing/routing guidance from the Simulink resource model, refreshed periodically; the SimEvents co-validation detail is collapsed behind a toggle by default

5.3 Central Backend Services

Ingestion API (dedup on `capture_id`) · Grading Pipeline Orchestrator (preprocessing → camera calibration → segmentation → dual-branch grading → confidence routing → rationale) · Referral & Notification Service (Twilio SMS, manual-follow-up on failure) · Admin Analytics Aggregator · Simulink Integration (scheduled resource-model run) · Grading Job Watchdog · MATLAB Session Supervisor · Auth Service · Audit Logging · Site & Camera Probation Tracking (a new PHC/camera family runs through mandatory review while its early outcomes are compared against the existing calibration baseline, before it graduates to normal tiering).

5.4 Database

PostgreSQL, with real migration tooling (`node-pg-migrate`). Core tables: `patients` , `cases` , `grading_results` (both branches' grades, agreement flag, confidence, uncertainty, conformal tier, claim state), `segmentation_outputs` (lesion counts per category, NV suspicion score, fovea reliability flag), `explainability_outputs` (Grad-CAM path, rationale report path, engine provenance), `ophthalmologist_reviews` , `referrals` , `phc_sites` , `users` , `access_log` .

5.5 API Contract

One contract (`docs/api-contracts.md`), consumed identically by both local front-ends. Core endpoints: `POST /api/v1/cases` (idempotent on `capture_id`), `POST /api/v1/cases/summary` (lightweight packet ahead of the image), chunked upload endpoints, `GET /api/v1/ophthalmologist/queue`, `POST /api/v1/cases/:id/review`, `GET /api/v1/phc/cases/:captureRef/report` (the PHC-scoped report a technician's own front-end reads back), `GET /api/v1/admin/*` (dashboard, system-health, referrals).

6. ML Pipeline

6.1 Image Quality Assessment (local, either front-end)

Classical CV heuristics — focus/blur, illumination, contrast, field-of-view, glare, motion artifact, eyelash occlusion, colour balance, border proportion — per-camera-family preset, implemented once in MATLAB and once as an equivalent JS reimplementations that both the desktop fallback tier and the entire mobile app run.

6.2 Preprocessing (central)

CLAHE, illumination normalization, denoising, Ben Graham-style circular crop — implemented exactly once, and that one implementation is what both trains and serves the classifier. A prior MATLAB reimplementations of this exact chain was removed after it was found to flip real grades on real images via a residual invisible on synthetic test tensors — a second implementation expected to agree with the first is treated as a standing risk, not a convenience.

6.3 Camera-Fingerprint Calibration (central)

Identifies the capture device family from vignetting, aspect ratio, and colour-channel gains, cross-checked against the worker-reported device field. A mismatch feeds the confidence-routing decision as a reason for mandatory review, not a pixel-level correction. New sites/camera families earn trust through mandatory review before graduating to normal tiering.

6.4 Optic Disc / Fovea Localization (central)

A U-Net regressor. Optic-disc localization: mean error 4.07px in 512-space, 98.7% within one disc radius (n=77). Fovea localization has a rare failure mode where the confidence heatmap has no real peak, silently rotating the quadrant axis used by the rule engine — a peak-confidence gate (threshold 0.37) catches this, marks the case `fovea_unreliable`, falls back to the plain image-axis convention, and forces mandatory review. Full numbers: `docs/ML_BENCHMARKS.md` §3.

6.5 Vessel Segmentation (central)

A U-Net trained on CHASE_DB1 (Dice 0.80 in-domain), complemented by a Frangi vesselness filter. An earlier "domain shift" diagnosis on cross-dataset performance was corrected: the true cause is an annotation-convention difference between datasets (different conventions mark thin vessels differently), not the model failing to generalize — though a genuinely lower cross-dataset Dice (0.62 on DRIVE) remains, and the live inference threshold is a fixed 0.5, not yet adjusted per-domain as originally specified. Feeds NV-suspicion scoring and false-positive suppression for microaneurysm detection.

6.6 Lesion Segmentation (central)

Two models: a **red-lesion model** for microaneurysms+haemorrhages, and a **hard-exudate model** (Dice 0.67, n=27) — "bright lesion" is exactly and only hard exudates. **The red-lesion model actually deployed is v2**, a 3-class architecture (background/MA/HE) with a class-specific minimum-size filter — a much lower size floor for microaneurysms, since most true microaneurysms are only a few pixels across and a filter tuned for haemorrhage-scale noise would discard most of them. v1 (a single combined-class predecessor) is the only version with its own measured Dice score (0.61); v2's own segmentation accuracy has not yet been independently re-measured, though it is confirmed live in production (real per-case, non-zero microaneurysm/haemorrhage counts in the database). As of 2026-09-30, v2 also runs through the MATLAB session by default like every other segmentation model (ONNX-imported to `red_lesion_unet_v2.mat`, parity-checked against its PyTorch source), rather than always running PyTorch — see `docs/ML_BENCHMARKS.md §3` for the full, honest breakdown of this gap. **Cotton-wool spots (soft exudates) are an explicit, disclosed scope exclusion** — training data for this specific lesion type was judged too sparse to support a real detector; the field is always reported as unmeasured, never as a false zero.

Neovascularization is a suspicion score (vessel density/branching/tortuosity near the optic disc), not a segmentation claim, because it has the least public training data of any lesion type here. Its own measured performance (AUC 0.29 on IDRiD, 0.38 on Messidor-2 — both below chance) means it is currently gated off and decides nothing; a CNN-predicted grade-4 triggers mandatory review on its own, independent of this score.

Standing rule: the lesion-count size filter and the rule engine's severity thresholds are one coupled, jointly-calibrated unit — changing one without recalibrating the other silently rescales what the thresholds mean.

6.7 DR Severity Classification — Two Independent Branches (central)

Branch A (CNN): EfficientNet-B0, transfer-learned with an ordinal-aware loss (grade confusions closer together on the severity scale count as smaller errors) plus class-weighted terms for severe-grade imbalance. A dedicated safety check runs alongside the argmax grade: if the combined probability of grade 3 and grade 4 exceeds 0.5, the case is marked referable regardless of the single most likely grade — because the most severe grade is also the hardest to resolve, and a near-miss on the top grade should never silently become a non-referral. **Currently deployed: `branchA_v2c`** (512×512 input), which replaced an earlier version specifically because that version's grade-4 (proliferative DR) recall was a ship-blocking 0.444 — on the official 103-image IDRiD test set, v2c's grade-4 recall is 13/13 (see `docs/ML_BENCHMARKS.md §1` for the full table and the caveats that come with it).

Branch B (rule engine): plain, testable code implementing the ICDR/ETDRS "4-2-1" rule directly on quadrant-mapped lesion counts — auditable against the clinical rule text line by line, not a black box. Falls back to the plain image-axis convention (skipping quadrant logic) whenever `fovea_unreliable` is set. Live, frozen thresholds: `RED_FLOOR=3`, `GRADE3_QUAD_MIN=3`, `RULE_MAX_GRADE=3` — not to be changed without an explicit decision (`CLAUDE.md`). A separate recalibration study exists but is explicitly marked "not read by any production code" and is not what ships.

Fusion: agreement between the branches is itself evidence supporting the case's confidence tier. Disagreement is an unconditional trigger for mandatory review with a required explicit resolution — never averaged away as noise. No system-wide agreement-rate statistic is currently reported (the only candidate data file is a small curated set of disagreement examples, not a representative sample — see `docs/ML_BENCHMARKS.md §4`).

6.8 Confidence Routing (central)

Every case's tier is the output of **one decision**, combining: temperature-scaled classifier confidence, Monte Carlo Dropout uncertainty, class-conditional ordinal conformal prediction, branch agreement (a disagreement is an unconditional override), camera-family mismatch / probation status, and capture-quality flags (including `fovea_unreliable`).

Tier	Meaning	Consequence
A — auto-clear	Singleton conformal set, high confidence, branch agreement, no probation/quality flag	Skips the ophthalmologist queue entirely
B — AI-assisted review	Narrow conformal interval	Full rationale report, target under 30s review
C — full manual review	Wide interval, disagreement, CNN grade-4, probation, or a quality flag	No shortcuts

A cross-fit safety study on the deployed model's calibration population (n=1161, 50 folds) found **zero grade-4 cases auto-cleared to Tier A** across 1000 fold-assignments, and a 0% false auto-clear rate for both referable and grade≥3 cases — the core safety property this tiering system exists to guarantee, holding with no exceptions found. Full numbers: docs/ML_BENCHMARKS.md §2.

6.9 Explainability (central)

Grad-CAM/Grad-CAM++ on the classifier's final convolutional layer, restricted to the retinal ROI. **Important display detail:** the Grad-CAM image is the model's own cropped working image, not the original capture — the crop can look tighter or offset from the photo shown elsewhere on the case, and both frontends now caption this explicitly rather than letting it read as a bug. Assembled, together with the four-category lesion evidence and which rule-engine criteria fired, into one per-case rationale. A trailing methodology caveat that the underlying evidence-summary text sometimes carries (e.g., a threshold being provisional) is now split into its own labeled "methodology note" on both the central and mobile UIs, rather than reading as additional clinical evidence.

6.10 Symptom + Risk Fusion (central)

Rule-based weighted scoring applied as a confidence adjustment after the image model's calibrated output — not a learned model, since no dataset currently links questionnaire data to outcomes.

6.11 Continual Learning (central, periodic)

Ophthalmologist overrides and their structured reasons are captured inside the same transaction as the review decision and exported (scripts/exportTrainingSet.js , with consent enforcement, dedupe, newest-label-per-case). **The scheduled fine-tuning job, validation gate, and promotion step are not implemented this round** — this is a stated scope boundary, not an oversight: corrections are captured and exportable; retraining is a manual, future-round step.

7. Systems Layer — Simulink Resource Model

A real, working discrete-event simulation (SimEvents), built as a script rather than hand-assembled in a GUI. Entities are patient images; the arrival process is the acquisition rate per PHC; network transmission is a bandwidth-constrained queue across good/poor/very-poor connectivity tiers; ophthalmologist review is a limited-capacity server (~30s Tier B, several minutes Tier C, with Tier C pre-empting Tier B on the shared reviewer pool). Outputs: queue length over time, average wait, bottleneck location, a plain-language resource recommendation — feeding the Admin Analytics Aggregator and Resource Recommendations screen. Bandwidth/timing parameters are modeled assumptions, not measured field data.

8. Resilience & Edge-Case Handling

- **Extended outages:** urgency-then-age sync ordering, storage-pressure warnings, a lightweight summary packet ahead of the full image under thin connectivity, PHC last-contact time feeding System Health, and a manual export-to-drive fallback for outages severe enough that even intermittent connectivity isn't achievable.
- **Ungradable images:** after a fixed number of failed retakes, the technician can mark a capture "best effort — proceed as ungradable," forcing mandatory Tier C review rather than an infinite retry loop or a silently dropped case.
- **Duplicate patients:** fuzzy name/age/phone matching at registration, technician confirms or merges before a new ID is created.
- **Per-eye capture:** every capture is tagged left/right; one visit can produce two independent graded cases.
- **Failed patient contact:** an undeliverable SMS flips the referral straight to manual-follow-up.
- **Collision-safe identity:** patient/capture IDs use PHC code + timestamp + random suffix, not an incrementing counter — a reset or reinstalled device can never regenerate an ID already in use centrally. Every case submission carries its `capture_id` as an idempotency key, enforced with a uniqueness constraint centrally.
- **Active system-health monitoring:** a silent PHC, a stuck grading job, a dead MATLAB session, and an unreviewed referable case are treated as the same class of failure — each with its own continuously-running check and threshold, surfaced on one consolidated screen, with plain-language summaries by default and raw detail behind a toggle.
- **Concurrent review claiming:** opening a case claims it; a second reviewer sees who holds it and cannot submit a conflicting decision while it's claimed.
- **Mandatory resolution on disagreement:** "Confirm" is never an available action when the two branches disagree — the reviewer must pick an explicit final grade.

9. Security (Current State)

`docs/system-design-v4.md` §11.1 states, as written, that this system has no authentication anywhere — this was accurate when written and is now **stale**, verified directly against the live, running code this session:

- **Central:** `AUTH_ENABLED=true`. Verified by logging in as both `district_admin` and `ophthalmologist` through the real `POST /api/v1/auth/login` (bcrypt-checked; a wrong password returns a genuine 401). The login screen's demo-account shortcut only exists when `DATA_MODE=mock`, which is not the default.
- **PHC desktop and mobile:** `LOCAL_AUTH_ENABLED=true` (set explicitly this integration pass — previously shipped `false` by default, which is the actual severity of the gap v4 was describing: a config default, not "auth doesn't exist"). Real bcrypt/session code, technician accounts provisioned via CLI (`npm run technician -- add <username> "<name>"`), password generated and printed once.
- **Mobile:** authenticates against the paired PC's real technician accounts over an encrypted peer channel (`/peer/login`), with an honest offline-cache fallback and a dev-only demo account gated to Expo Go builds — never a silent success fabrication on failure.
- **Encryption:** AES-256-GCM at rest for stored images, Grad-CAM overlays, and report PDFs (`MEDIA_ENCRYPTION_KEY`); TLS supported for both backends.
- **Audit logging:** every access to patient data is recorded (`access_log` table).

What remains a real, stated gap, not resolved by the above: this is a prototype-stage security floor, not a production compliance claim. Role-based access control is enforced server-side on central's endpoints; a full security audit (rate limiting, secret rotation, penetration testing) has not been performed.

10. Honest Limitations (Stated on Purpose)

- **This is a screening decision-support system, not an autonomous diagnostic** — every positive result is confirmed by an ophthalmologist before it reaches a patient.
- **Rural deployment without a MATLAB license is not solved this round.** The persistent MATLAB session serving Branch A requires a MATLAB license and the Deep Learning Toolbox on whatever machine runs it. The local quality gate solves this exact problem for itself (MATLAB Compiler + free Runtime); the same packaging has not yet been applied to the central inference path.
- **Specificity (82.1%) is below the >85% target** even though sensitivity clears its >90% target — meaning only ~4% of cases fully auto-clear at the current calibration; most cases still reach a human reviewer.
- **The red-lesion model actually running in production (v2) has no independently measured accuracy score of its own** — only its v1 predecessor's does (a different, single-class architecture). See `docs/ML_BENCHMARKS.md` §3.
- **Neovascularization detection is measured and has failed** (below-chance AUC) — it is gated off, a distinct fact from the grade-4 recall issue, which is resolved on the deployed model.
- **Diabetic macular edema cannot be reliably ruled out from color fundus photographs alone.**
- **The grading queue is in-memory** and only recovers stuck jobs at server restart plus the Grading Job Watchdog's periodic scan.
- **Automated test coverage is real but partial** — mobile (28 unit tests) and PHC backend (41 tests) pass; central backend has no automated suite, relying on targeted manual verification scripts.
- **Simulink parameters are modeled assumptions, not measured field data.**
- **Messidor-2 was used for model selection between candidate classifier versions, not as an untouched external validation set** — so it does not substitute for one.
- **The "integrated pipeline beats a single technique" claim is not statistically supported yet** at matched coverage.

11. Datasets

Dataset	Size	Used for
APTOS 2019	3,662 images, 5-class DR grade	Branch A classifier training
IDRiD	81 (segmentation) + 516 (grading) + 516 (localization)	Segmentation, classification, localization, and the official held-out test evaluation
CHASE_DB1	Standard vessel-segmentation set	Vessel segmentation training
Messidor-2	~1,748 images	Model selection across classifier candidates; external generalization check

Only public datasets are used anywhere in this system — no real patient data, per project policy.

12. Where to Look for More

- **Full ML numbers, with sources and reproduction commands:** `docs/ML_BENCHMARKS.md`
- **API request/response shapes:** `docs/api-contracts.md`
- **Setup and running the system:** `README.md`

- **Demo walkthrough (scene by scene):** docs/DEMO_RUNBOOK.md ; condensed video script: docs/DEMO_SCRIPT.md
- **What's been fixed vs. what v4 still describes accurately:** docs/STALE_CLAIMS_AUDIT.md
- **Original design rationale, in full:** docs/system-design-v4.md